Geometry of Rényi Entropy on the Majorization Lattice

Anuj Yadav
EPFL

Joint work with Yanina Shkel





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Outline

1. Majorization Lattice
2. Couplings on the Majorization Lattice
3. Subadditivity
4. Supermodularity
5. Applications

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Majorization

- Ordered PMFs $\mathcal{P}_n$: $\mathbf{p} = (p_1, \ldots, p_n)$, $p_1 \geq \cdots \geq p_n$.
- **Definition: q** *majorizes* **p**, $\mathbf{p} \preceq \mathbf{q}$, if

$$\sum_{i=1}^{k} p_i \leq \sum_{i=1}^{k} q_i \qquad \forall\, k \in [n].$$

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Applications in: econometrics, entanglement, couplings, etc.

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Rényi Entropy and Majorization

- Rényi entropy of order $\alpha \in [0, \infty]$:

$$H_\alpha(\mathbf{p}) = \frac{1}{1-\alpha} \log \sum_{i=1}^{n} p_i^\alpha$$

  - $H_0(\mathbf{p}) = \log |\mathrm{supp}(\mathbf{p})|$ (max-entropy);
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- **Schur concavity:** more concentrated $\Rightarrow$ lower entropy,

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- **Aggregation:** if $\mathbf{q}$ is obtained by merging masses of $\mathbf{p}$ ($\mathbf{p} \sqsubseteq \mathbf{q}$), then $\mathbf{p} \preceq \mathbf{q}$.

  e.g. $(0.5, 0.3, 0.2) \sqsubseteq (0.5, 0.5)$ - "aggregation implies majorization".

The Majorization Lattice

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- **Complete:** *every* subset $\mathcal{S} \subseteq \mathcal{P}_n$ has
  - a glb $\bigwedge \mathcal{S}$: the *largest* PMF majorized by every element of $\mathcal{S}$,
  - a lub $\bigvee \mathcal{S}$: the *smallest* PMF majorizing every element of $\mathcal{S}$.
  - Both $\bigwedge \mathcal{S} \in \mathcal{P}_n$ and $\bigvee \mathcal{S} \in \mathcal{P}_n$.

For a pair: $\mathbf{p} \wedge \mathbf{q} \preceq \mathbf{p}$, $\mathbf{q} \preceq \mathbf{p} \vee \mathbf{q}$.

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  For a pair: $\mathbf{p} \wedge \mathbf{q} \preceq \mathbf{p}$, $\mathbf{q} \preceq \mathbf{p} \vee \mathbf{q}$.
- **Computable** from partial sums: for $\mathcal{S} = \{\mathbf{p}^{(j)}\}_{j=1}^{m}$,
  - glb: $\sum_{i \le k} \left(\bigwedge \mathcal{S}\right)_i = \min_j \sum_{i \le k} p_i^{(j)}, \quad k \in [n]$,
  - lub: $\max_j \sum_{i \le k} p_i^{(j)}$, then average violating blocks (via least concave majorant (LCM) operation)

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Entropy on Lattices

- **Boolean lattice** (subsets of random variables): Shannon entropy is *submodular* — polymatroids, Shannon-type inequalities.
- **Majorization lattice** [Cicalese–Vaccaro '02]: Shannon entropy is

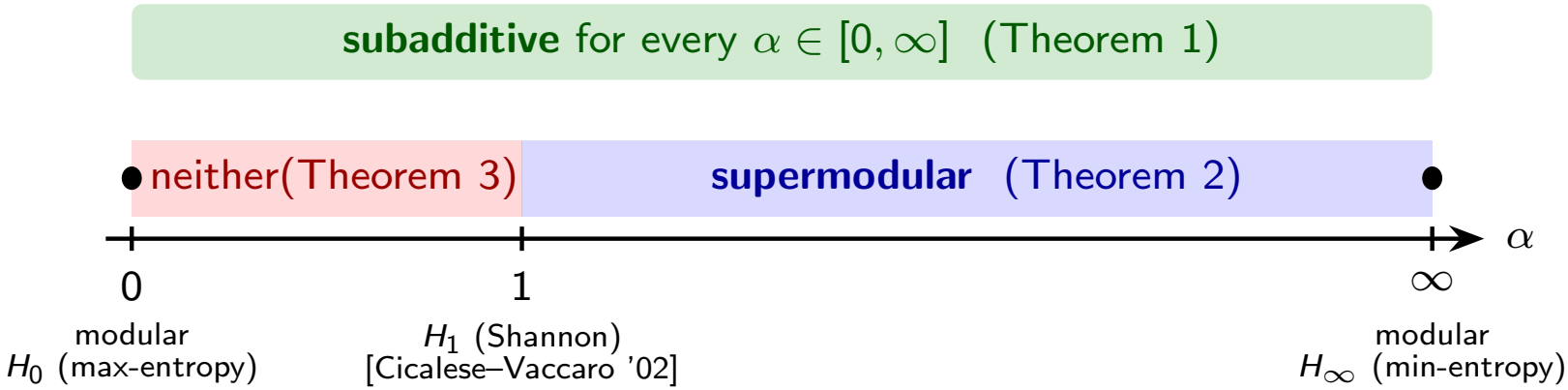
$$\begin{aligned} \text{subadditive}: &\quad H(\mathbf{p} \wedge \mathbf{q}) \;\leq\; H(\mathbf{p}) + H(\mathbf{q}), \\ \text{supermodular}: &\quad H(\mathbf{p}) + H(\mathbf{q}) \;\leq\; H(\mathbf{p} \wedge \mathbf{q}) + H(\mathbf{p} \vee \mathbf{q}). \end{aligned}$$

$\rightsquigarrow$ an "information-theoretic calculus" on the lattice.

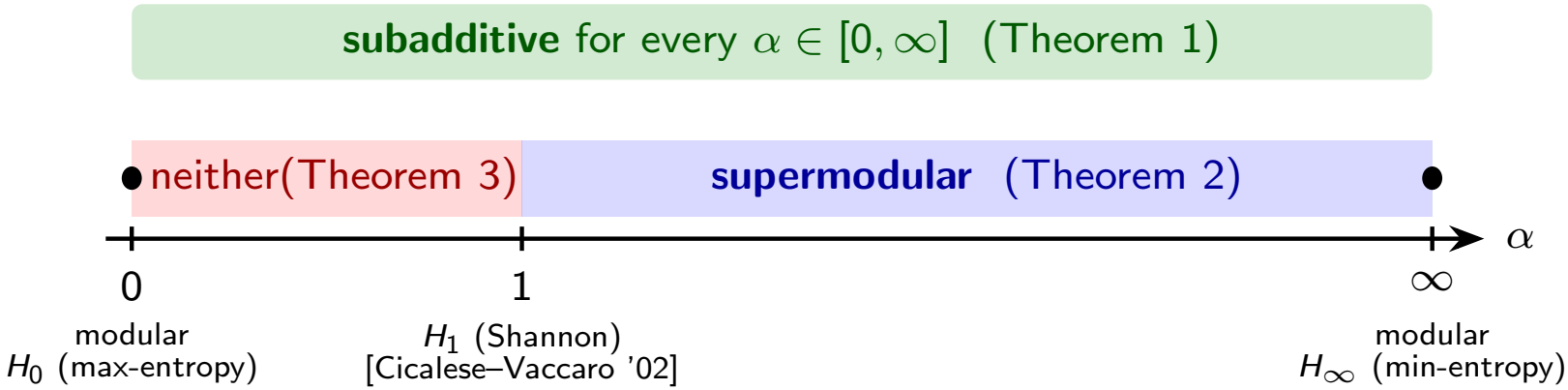
What can we say about the Rényi entropy of every order α ?

F. Cicalese and U. Vaccaro, "Supermodularity and subadditivity properties of the entropy on the majorization lattice," IEEE Trans. IT, vol. 48, no. 4, pp. 933–938, 2002.

The Answer

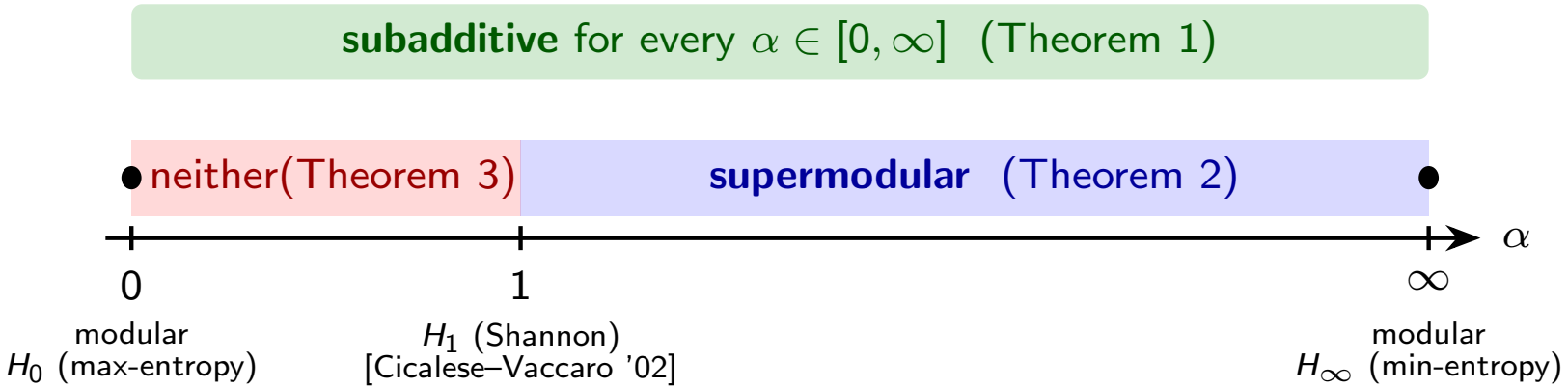

subadditive for every $\alpha \in [0, \infty]$ (Theorem 1)
neither(Theorem 3)
supermodular (Theorem 2)
α
0
1
∞
modular
H_0 (max-entropy)
H_1 (Shannon)
[Cicalese–Vaccaro '02]
modular
H_∞ (min-entropy)

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- **Subadditivity:** via coupling methods.

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- **Subadditivity:** via coupling methods.
- **Supermodularity:** via geometry of Lorenz curves.

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Two Canonical Couplings

Given $\mathbf{p}, \mathbf{q} \in \mathcal{P}_n$ with cumulative sums $P_i = \sum_{k \le i} p_k$, $Q_j = \sum_{k \le j} q_k$:

- **Independent coupling:** $\pi_I(\mathbf{p}, \mathbf{q})(i, j) = p_i \, q_j$.
- **Comonotone (north-west) coupling:**

$$\pi_c(\mathbf{p}, \mathbf{q})(i, j) = \left| (P_{i-1}, P_i] \cap (Q_{j-1}, Q_j] \right|$$

independent π_I

	0.6	0.2	0.2
0.5	0.30	0.10	0.10
0.4	0.24	0.08	0.08
0.1	0.06	0.02	0.02

comonotone π_c

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0.1	0	0	0.1

$\mathbf{p} = (0.5, 0.4, 0.1)$, $\mathbf{q} = (0.6, 0.2, 0.2)$; $\pi^{\downarrow}$: masses of π in non-increasing order.

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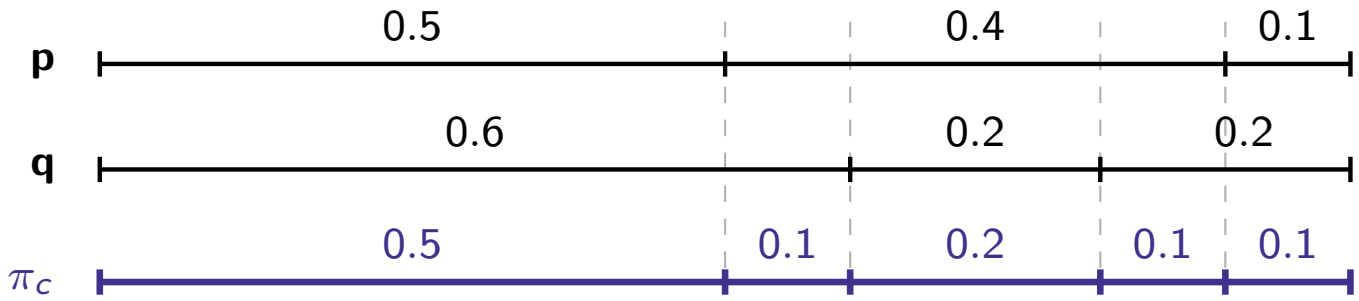
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- π_I: max. Shannon-entropy coupling; π_c: monotone staircase, $\le 2n - 1$ atoms.

Comonotone Coupling constructs the Majorization Lattice

Lemma 1: $\mathbf{p} \wedge \mathbf{q}$ is an *aggregation* of $\pi_c(\mathbf{p}, \mathbf{q})$, and $\mathbf{p} \vee \mathbf{q}$ is the LCM of an aggregation of $\pi_c(\mathbf{p}, \mathbf{q})$.

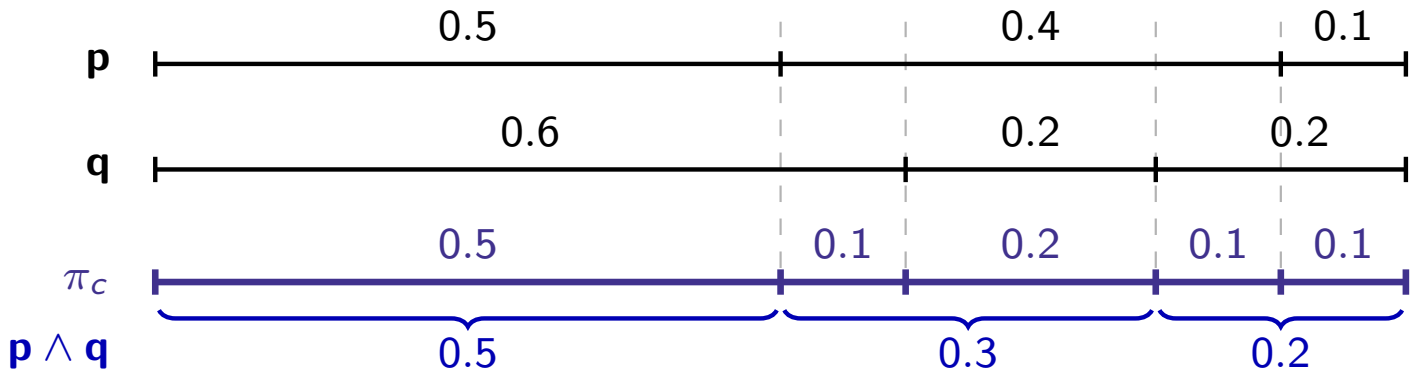


- π_c: gaps between the *merged* breakpoints of $\mathbf{p}$ and $\mathbf{q}$.

A. K. Yadav and Y. Y. Shkel, "Couplings and entropic inequalities on the majorization lattice," EPFL Infoscience preprint, 2026.

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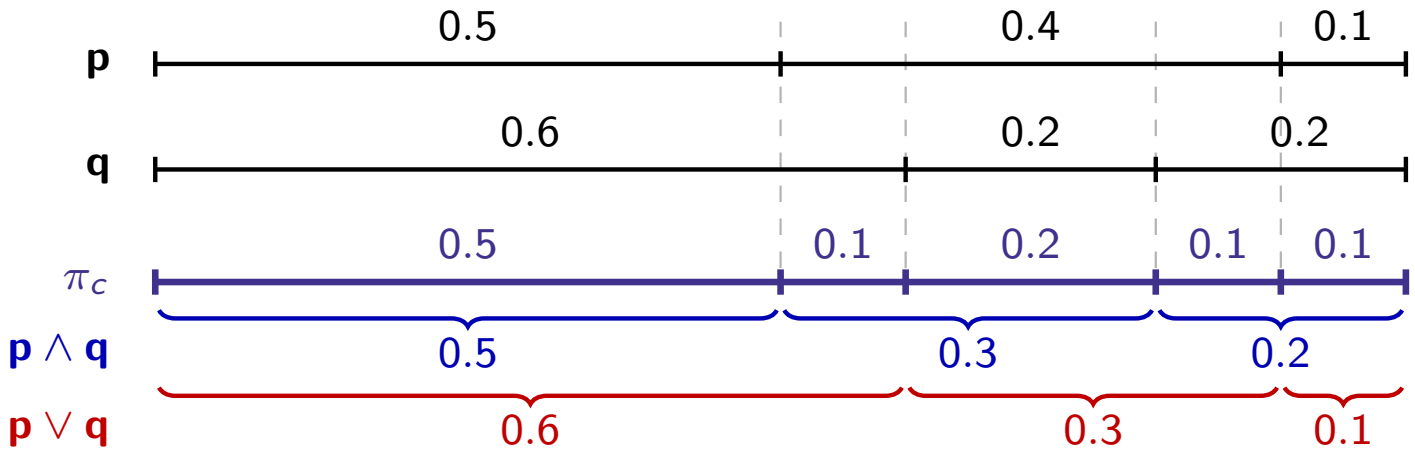


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- π_c: gaps between the *merged* breakpoints of $\mathbf{p}$ and $\mathbf{q}$.
- glb: cut at $\min\{P_k, Q_k\}$; lub: cut at $\max\{P_k, Q_k\}$ (+ LCM).
- Consequence: $\pi_c^{\downarrow}(\mathbf{p}, \mathbf{q}) \preceq \mathbf{p} \wedge \mathbf{q}$.

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Order of Majorization

Lemma 2: For all $\mathbf{p}, \mathbf{q} \in \mathcal{P}_n$,

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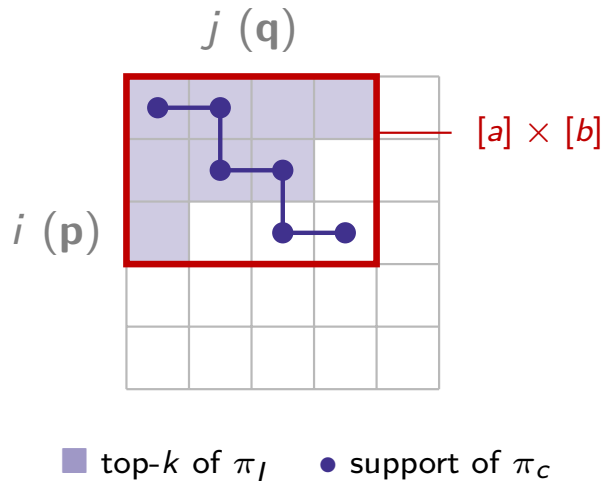
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Proof sketch:

- top-k masses of π_I: a Young diagram inside a minimal north-west box $[a] \times [b]$, total mass $\leq P_a Q_b \leq \min\{P_a, Q_b\}$.
- π_c puts *exactly* $\min\{P_a, Q_b\}$ in $[a] \times [b]$, on $\leq a + b - 1 \leq k$ atoms.
- $\Rightarrow$ every top-k sum of $\pi_c^{\downarrow}$ dominates that of $\pi_I^{\downarrow}$.



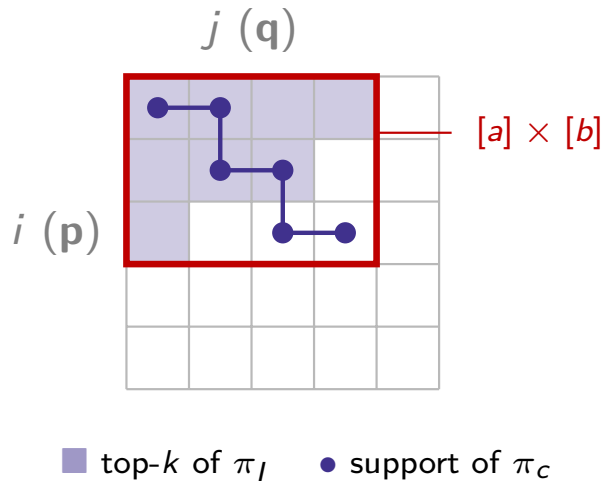
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Extends to any m marginals: $\pi_I^{\downarrow}(\mathcal{S}) \preceq \pi_c^{\downarrow}(\mathcal{S})$, $\mathcal{S} = \{\mathbf{p}^{(i)}\}_{i=1}^{m}$.

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Subadditivity of Rényi Entropy

Theorem 1: For every $\alpha \in [0, \infty]$ and all $\mathbf{p}, \mathbf{q} \in \mathcal{P}_n$,

$$H_\alpha(\mathbf{p} \wedge \mathbf{q}) \;\leq\; H_\alpha(\mathbf{p}) + H_\alpha(\mathbf{q})$$

with equality iff $\mathbf{p}$ or $\mathbf{q}$ is deterministic.

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Proof: chain the two lemmas together,

$$\pi_I^{\downarrow}(\mathbf{p}, \mathbf{q}) \underset{\text{Lemma 2}}{\preceq} \pi_C^{\downarrow}(\mathbf{p}, \mathbf{q}) \underset{\text{Lemma 1}}{\preceq} \mathbf{p} \wedge \mathbf{q}$$

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Then, from Schur concavity of H_α,

$$H_\alpha(\mathbf{p} \wedge \mathbf{q}) \;\leq\; H_\alpha(\pi_c) \;\leq\; H_\alpha(\pi_I) \;=\; H_\alpha(\mathbf{p}) + H_\alpha(\mathbf{q}).$$

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The comonotone coupling sits *between* the glb and the independent coupling.

Consequences

- **Sandwich:** since $\mathbf{p} \wedge \mathbf{q} \preceq \mathbf{p}, \mathbf{q}$,

$$H_\alpha(\mathbf{p} \wedge \mathbf{q}) \leq H_\alpha(\mathbf{p}) + H_\alpha(\mathbf{q}) \leq 2H_\alpha(\mathbf{p} \wedge \mathbf{q})$$

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- m **marginals:** for $\{\mathbf{p}^{(i)}\}_{i=1}^m \subset \mathcal{P}_n$,

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α	$H_\alpha(\mathbf{p} \wedge \mathbf{q})$	$H_\alpha(\mathbf{p}) + H_\alpha(\mathbf{q})$	$2H_\alpha(\mathbf{p} \wedge \mathbf{q})$
0.5	1.547	2.993	3.095
1	1.513	2.829	3.026
2	1.454	2.562	2.908

(in bits); $\mathbf{p} = (0.6, 0.2, 0.2)$, $\mathbf{q} = (0.45, 0.4, 0.15)$.

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Supermodularity of Rényi Entropy

Definition: A function f is *supermodular* on a lattice if $\forall\ x, y$,

$$f(x) + f(y) \leq f(x \wedge y) + f(x \vee y).$$

Theorem 2: For every $\alpha \in \{0\} \cup [1, \infty]$ and all $\mathbf{p}, \mathbf{q} \in \mathcal{P}_n$,

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- For $\alpha \in \{0, \infty\}$: **equality** holds, i.e., H_0, H_∞ are *modular*.

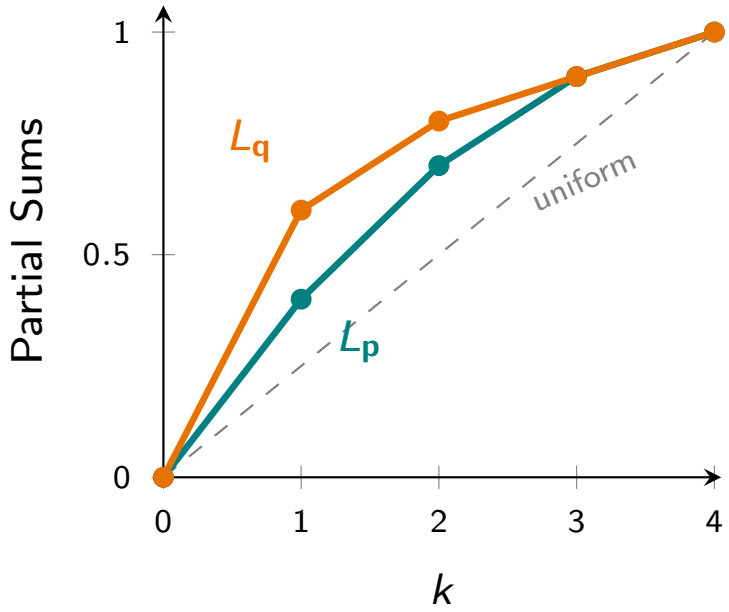
$$(\mathbf{p} \wedge \mathbf{q})_1 = \min\{p_1, q_1\}, \quad (\mathbf{p} \vee \mathbf{q})_1 = \max\{p_1, q_1\}.$$

Lorenz Curves

- **Lorenz curve** of $\mathbf{p} \in \mathcal{P}_n$: the piecewise-linear interpolation of the points

$$\left(k,\ \sum_{i \le k} p_i\right), \quad k = 0, \ldots, n.$$

- Slope on $[k-1, k]$ is p_k.
  $\implies$ $L_{\mathbf{p}}$ is **concave**.



$\mathbf{p} = (0.4, 0.3, 0.2, 0.1)$

$\mathbf{q} = (0.6, 0.2, 0.1, 0.1)$

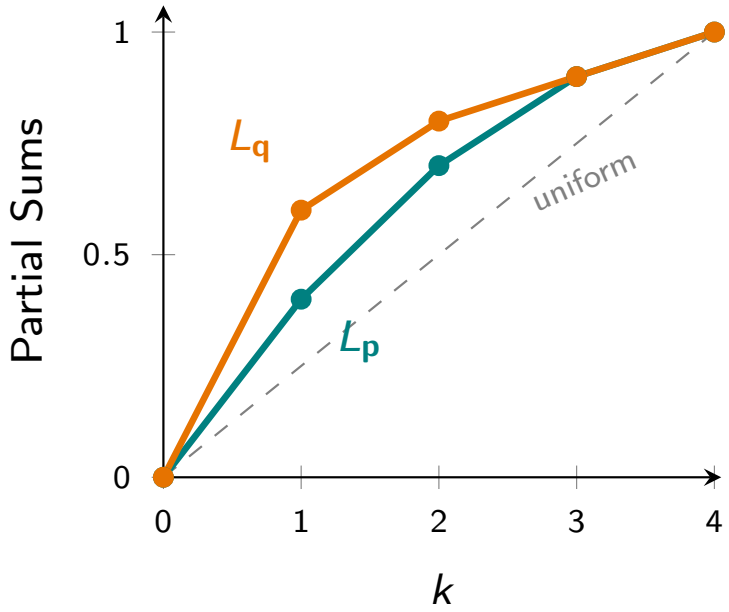
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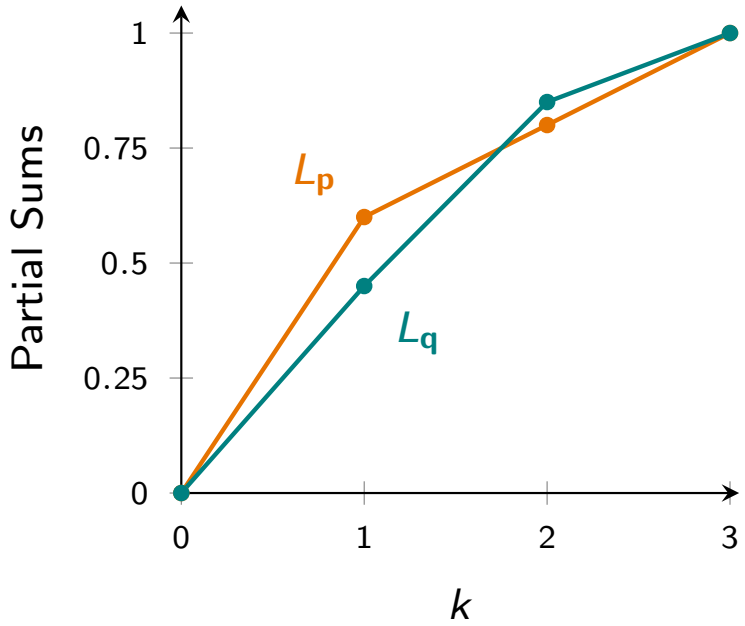


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GLB and LUB via Lorenz Curves

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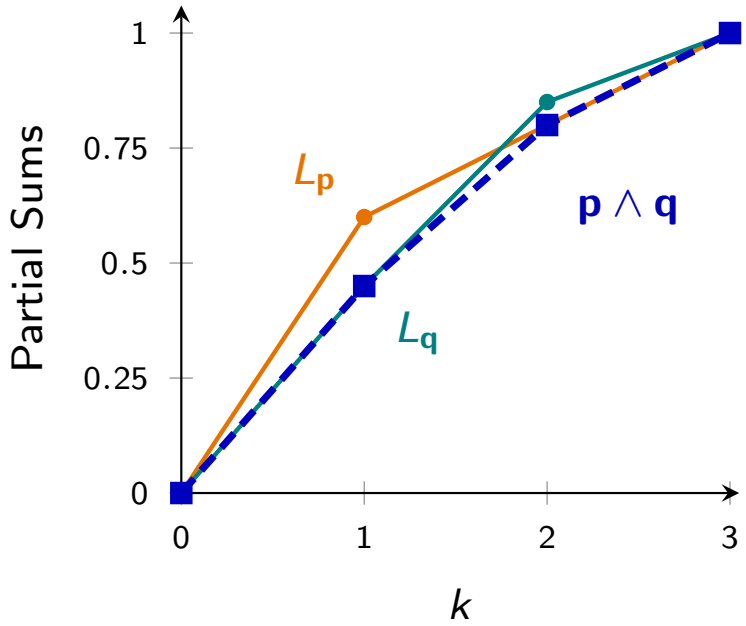
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R. Bapat, "Majorization and singular values III," Linear Algebra Appl., 1991;
G. M. Bosyk et al., "Optimal common resource in majorization-based resource theories," New J. Phys., vol. 21, 083028, 2019.

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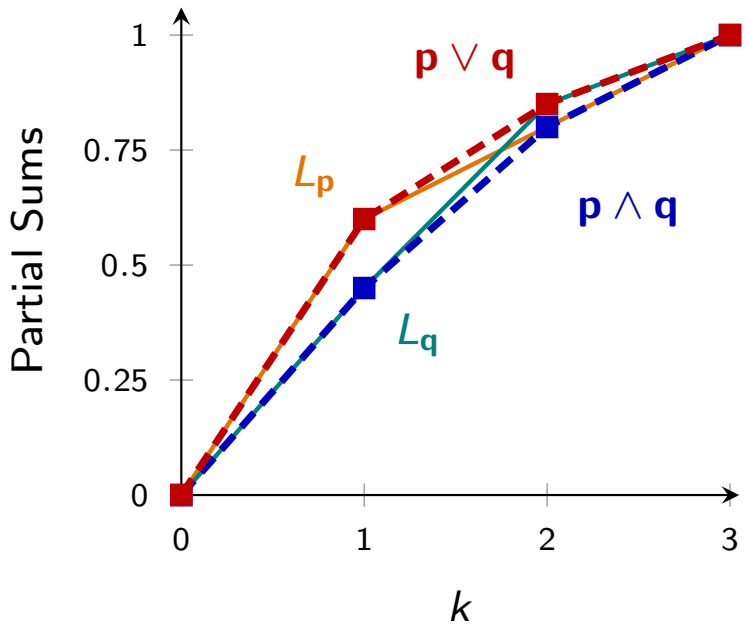
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- **lub:** β with partial sums $\max\{L_{\mathbf{p}}(k), L_{\mathbf{q}}(k)\}$; if $\beta \notin \mathcal{P}_n$, average violating blocks: $\mathbf{p} \vee \mathbf{q} = \mathrm{LCM}(\beta)$ (least concave majorant).



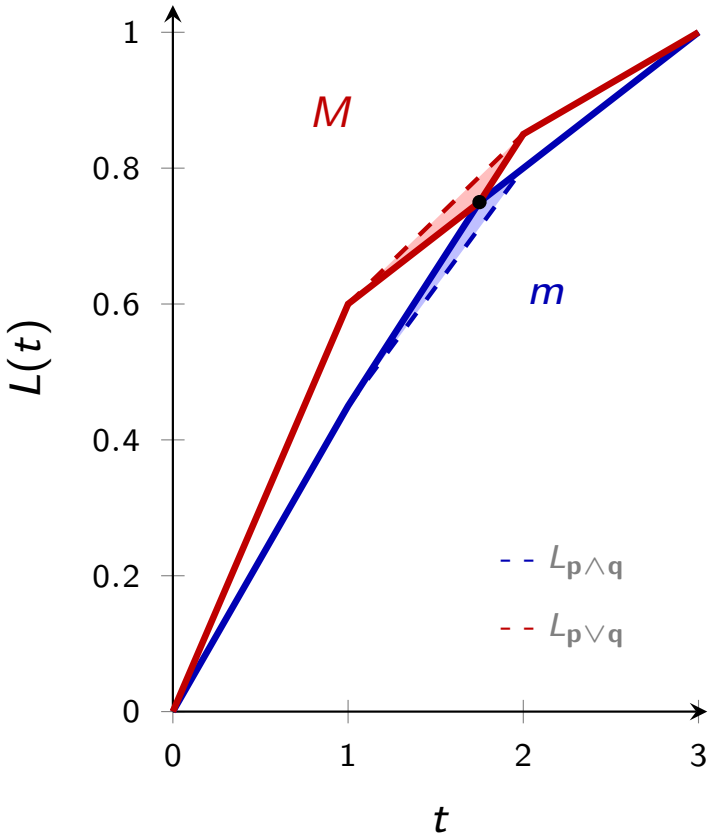
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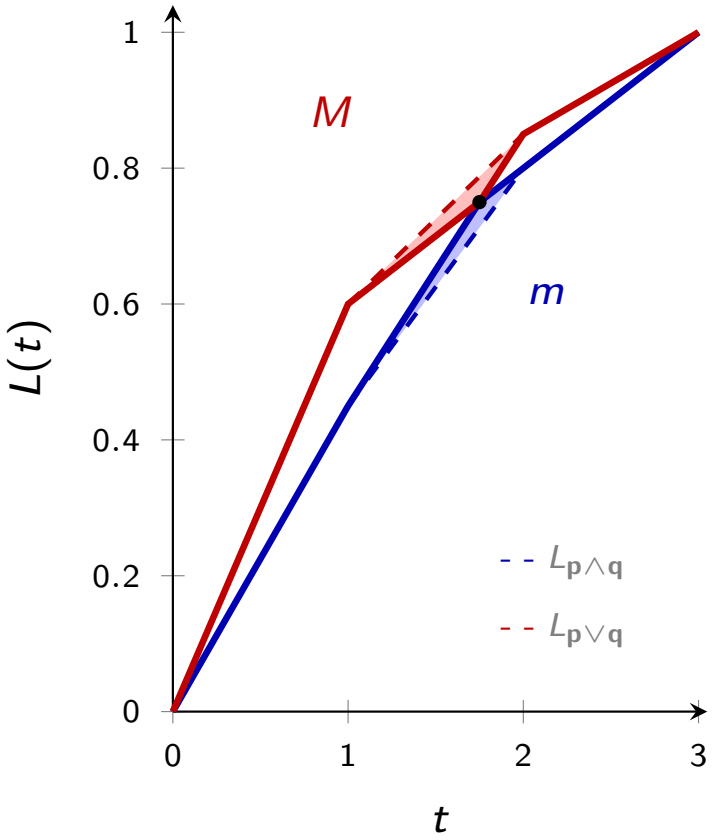
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Proof Idea: via Lorenz Geometry ($\alpha \geq 1$)



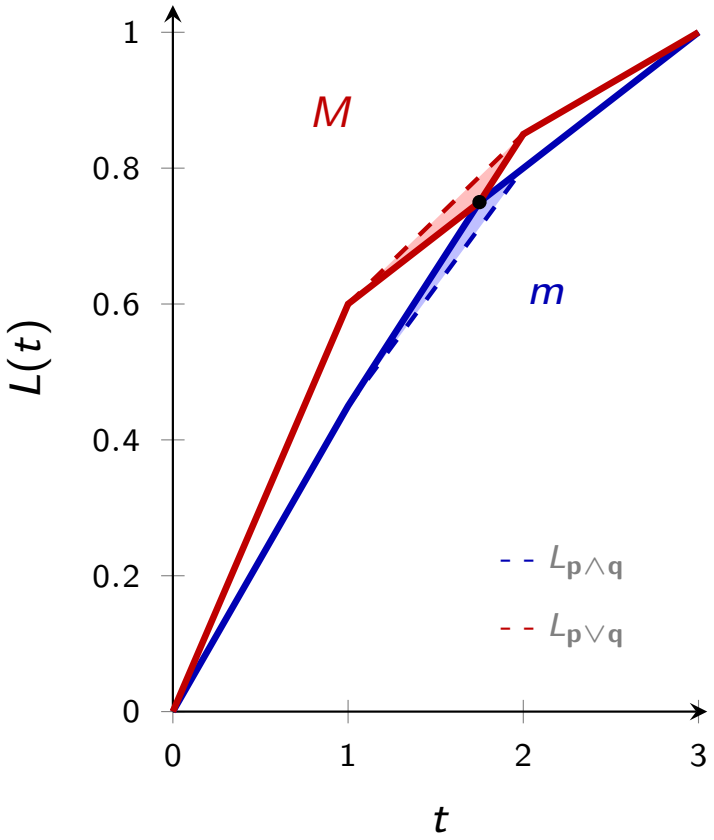
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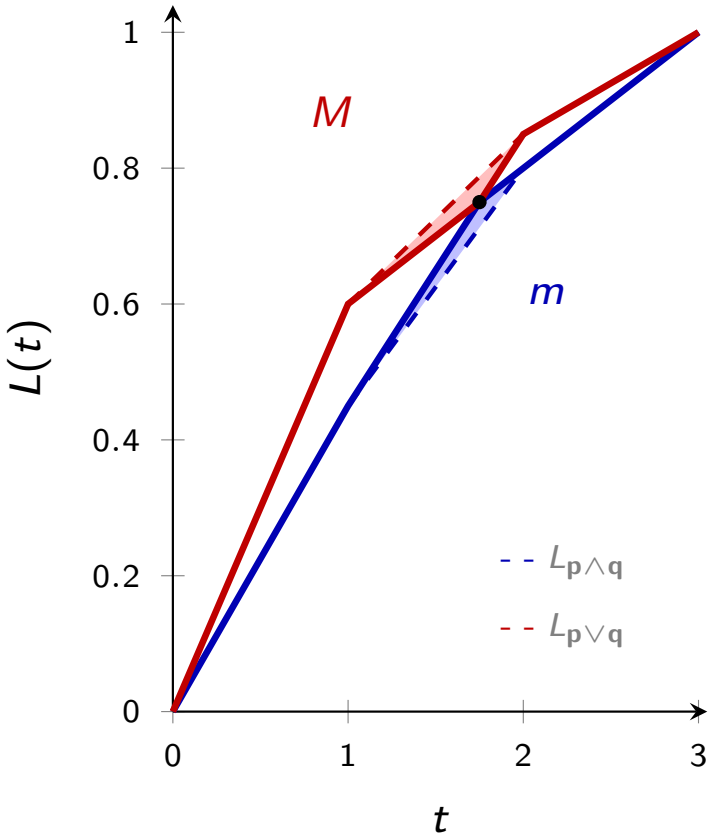
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3. $L_{\mathbf{p}\wedge\mathbf{q}}$, $L_{\mathbf{p}\vee\mathbf{q}}$ are **chords** of m, M ($+$ LCM); Jensen ($u \mapsto u^\alpha$ convex): $S_\alpha(\mathbf{p}\wedge\mathbf{q}) + S_\alpha(\mathbf{p}\vee\mathbf{q}) \leq S_\alpha(\mathbf{p}) + S_\alpha(\mathbf{q})$.

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4. Schur convexity of S_α orders the four terms $\Rightarrow$ $S_\alpha(\mathbf{p})S_\alpha(\mathbf{q}) \geq S_\alpha(\mathbf{p}\wedge\mathbf{q})S_\alpha(\mathbf{p}\vee\mathbf{q})$.

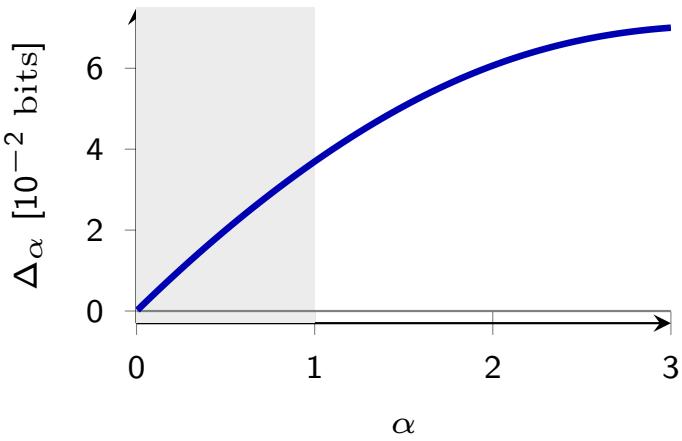
Apply $\frac{1}{1-\alpha}\log(\cdot)$ (negative factor for $\alpha > 1$) $\Rightarrow$ supermodularity.

The Regime $\alpha \in (0, 1)$

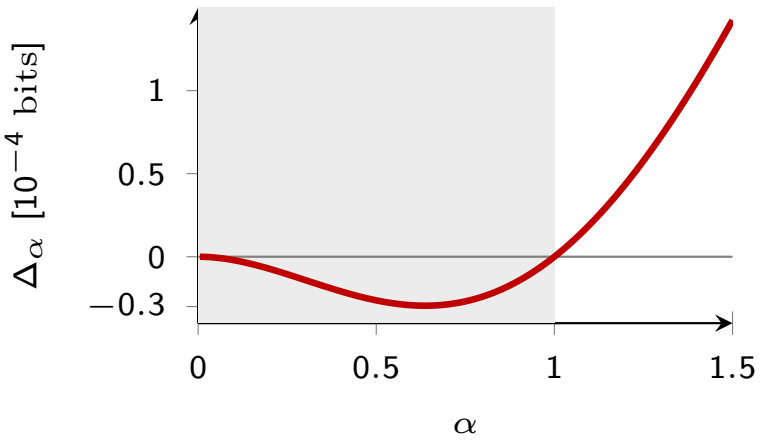
Theorem 3: For $n \geq 3$ and every $\alpha \in (0, 1)$, H_α is **neither** supermodular **nor** submodular.

Gap: $\Delta_\alpha(\mathbf{p}, \mathbf{q}) := H_\alpha(\mathbf{p} \wedge \mathbf{q}) + H_\alpha(\mathbf{p} \vee \mathbf{q}) - H_\alpha(\mathbf{p}) - H_\alpha(\mathbf{q})$.

Example 1: $\Delta_\alpha > 0$



Example 2: $\Delta_\alpha < 0$ on $(0, 1)$



Ex. 1: $\mathbf{p} = (0.6, 0.2, 0.2)$, $\mathbf{q} = (0.45, 0.4, 0.15)$; Ex. 2: $\mathbf{p} = (0.42, 0.28, 0.2, 0.1)$, $\mathbf{q} = (0.4, 0.3, 0.22, 0.08)$.

Outline

1 Majorization Lattice

2 Couplings on the Majorization Lattice

3 Subadditivity

4 Supermodularity

5 Applications

Minimum Rényi Entropy Couplings

Given marginals $\mathcal{S} = \{\mathbf{p}^{(i)}\}_{i=1}^{m}$, find the coupling of least Rényi entropy:

$$H_\alpha^\star(\mathcal{S}) := \min_{\pi \in \mathcal{C}(\mathcal{S})} H_\alpha(\pi) \qquad \text{(NP-hard)}.$$

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- **Optimal for** $\alpha = \infty$**:** largest atom $\min_i p_1^{(i)}$, the largest possible.

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An Entropic Distance on Ordered PMFs

Definition: for $\mathbf{x}, \mathbf{y} \in \mathcal{P}_n$,

$$d_\alpha(\mathbf{x}, \mathbf{y}) := H_\alpha(\mathbf{x}) + H_\alpha(\mathbf{y}) - 2\, H_\alpha(\mathbf{x} \vee \mathbf{y})$$

Theorem: d_α is a **metric** on $\mathcal{P}_n$ for every $\alpha \in [1, \infty)$.

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- **Econometrics:** the distance to the uniform $\mathbf{u}_n$ is a transform of the generalized entropy (inequality) index GE_α (Theil index at $\alpha = 1$):

$$\begin{aligned} d_\alpha(\mathbf{x}, \mathbf{u}_n) &= \log n - H_\alpha(\mathbf{x}) \\ &= \tfrac{1}{\alpha - 1} \log\left(1 + \alpha(\alpha - 1)\, \mathrm{GE}_\alpha\right) \\ &= D_\alpha(\mathbf{x} \| \mathbf{u}_n). \end{aligned}$$

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Entropic Optimal Transport: Cost of Regularization

For a cost matrix C, OT plan $\pi^\star \in \arg\min_{\pi \in \mathcal{C}(\mathbf{p},\mathbf{q})} \langle C, \pi \rangle$; entropic (Sinkhorn) plan, $\varepsilon > 0$:

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- Never worse than $\varepsilon \min\{H(\mathbf{p}), H(\mathbf{q})\}$ (as $\mathbf{p} \wedge \mathbf{q} \preceq \mathbf{p}, \mathbf{q}$); strictly better whenever $\mathbf{p}$ and $\mathbf{q}$ are incomparable.

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Summary

- The glb and lub of the majorization lattice are aggregations of the **comonotone coupling**.
- **Comonotone coupling majorizes independent coupling:** $\pi_I^{\downarrow} \preceq \pi_C^{\downarrow}$ (any number of marginals).
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Thank You!